

Explain how new evidence or different perspectives can lead to changes in scientific knowledge

Learning intention

- explain how new evidence or different perspectives can lead to changes in scientific knowledge.

Teach and model

- researching why Dmitri Mendeleev developed a different representation of the periodic table.
- considering how advances in technologies have enabled the repair and replacement of organs using synthetic materials.
- examining the evidence that led to the acceptance of the theory of plate tectonics over the idea of continental drift.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.